

Gene Boo Ee Jin

Quantitative Finance, Mathematics & Computational Systems Architect

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📦 3D Portfolio

⚡ Professional Summary

Quantitative finance and computational systems practitioner with 25+ years across derivatives, risk analytics, numerical methods, model validation, balance sheet risk, software engineering and creative computing. Current work spans IRRBB behavioural modelling, SA-CCR interpretation, structured-product valuation optimisation, local-volatility implementation, Python/SQL automation, OCR-enabled reporting workflows and model-risk defensibility. Broader portfolio work includes cloud-hosted analytics, WebGL/Three.js visualisation, local LLM experimentation, technical writing and mathematical edutainment tools.

📦 Core Competencies

Quantitative Finance

- ▶ Derivatives Pricing: FFT, Quadrature, MGFs, Local Vol, SABR, FD, Variance-Reduced MC
- ▶ Autocallables, FXO/IRO, Equity, Commodity and Energy Derivatives
- ▶ Risk Metrics: VaR, ES, PFCE, Euler Allocation
- ▶ SA-CCR, Risk-Neutral Valuation, PCA/PCR and Factor Models

Technical Engineering

- ▶ Python: NumPy, SciPy, pandas, Polars, mpmath, DuckDB, joblib
- ▶ VBA/Excel, C++ Quant Libraries, SQL Server/T-SQL, MySQL
- ▶ REST APIs, Postman, Grafana, Salesforce, Google Cloud Run
- ▶ HTML, CSS, JavaScript, WebGL, Three.js, YAML, Parquet

Risk & Governance

- ▶ Market, Operational, Liquidity and Banking Book Risk
- ▶ IRRBB / EVE / CA-SA NMD / TDRR Behavioural Modelling
- ▶ Model Validation for Traded and Non-Traded Portfolios
- ▶ Regulatory Contexts: BNM, MAS, HKMA, Basel

Automation, AI & Creative Computing

- ▶ OCR/PDF table extraction, Power Query and reporting automation
- ▶ ChatGPT, Copilot, Gemini, Grok, DeepSeek, Kimi, Ollama, Hugging Face
- ▶ NL-to-PRQL/SQL local LLM experimentation for Parquet Viewer
- ▶ Fourier epicycles, audio-reactive systems, DSP and mathematical edutainment

🧰 Professional Experience

IBMR Projects Consultant Ambank Group

Mar 2025 – Present

- ▶ Implemented BNM-aligned TDRR behavioural modelling for term deposits, covering a base cohort model and two challenger smoothing approaches using EWMA and tail-smoothed redemption curves.
- ▶ Implemented CA-SA/NMD behavioural modelling using cohort-based best-choice scalar yield-curve pillar PTR against market rates, plus multivariate PCA-PTR regression across the full yield curve to capture broader repricing behaviour while reducing multicollinearity-related instability; also explored cohort-based challenger modelling using Logistic Regression and XGBoost.
- ▶ Corrected and optimised an autocallable structured-product valuation module, upgrading the NumPy implementation with disk caching and vectorised antithetic/control-variate logspace Monte Carlo routines; reduced convergence requirement from approximately 1,000,000 paths to 15,000 paths and single-contract runtime from around 3 minutes to 40 ms.
- ▶ Corrected and vectorised an existing NumPy Dupire local-volatility implementation, reducing L-BFGS solver time from approximately 13 minutes to 4 seconds for a grid of five local-volatility pillar estimations.
- ▶ Implemented automation scripts for annual-report PDF table extraction across major Malaysian peer banks using OCR and Power Query techniques, improving access to risk-reporting fields and numerics and reducing peer-comparison turnaround from approximately 2 days to 2 hours.
- ▶ Provided SA-CCR implementation guidance for structures that did not fit neatly within vanilla framework interpretation, and supported prepayment/OOC early-withdrawal analytics using Python, Polars, SQL and large historical datasets.

 Python / NumPy / SciPy / pandas / Polars

 SQL Server / DuckDB / Parquet

 Excel / Power Query / VBA

 OCR / PDF Extraction

Product Specialist Qontigo / AxiomaRisk (SimCorp)

Dec 2021 – Jun 2023

- ▶ Delivered quantitative multi-asset-class product support for AxiomaRisk and performance attribution across buy-side institutional workflows.
- ▶ Conducted pre- and post-sales demonstrations showcasing AxiomaRisk functionality to existing and potential clients, covering VaR, ES, liquidity analytics, factor models, PCA modes and multi-asset risk outputs.
- ▶ Built Python/API automation using AxiomaRisk REST API, cURL, requests and Postman for client and internal training workflows, including position loading, report extraction, validation and Excel analytics.
- ▶ Built external model validation spreadsheets and Jupyter notebooks explaining selected AxiomaRisk computations including VaR, structured-product pricing, Euler decomposition and risk attribution analytics.
- ▶ Worked with Kubernetes monitoring tools including Grafana and Kibana, development tools such as Jira and sales tooling such as Salesforce.

 Python / Pandas / XLWings

 REST API

 Postman / Grafana / Kibana

 Salesforce / Jira

Senior Quantitative Analyst Maybank Group

Sep 2019 – Aug 2021

- ▶ Developed Quasi Monte Carlo-bootstrap operational-risk (loss-distribution building using severity and frequency curves using ORX external data combined with internal loss-event database containing losses, near-misses etc) capital frameworks using empirical + Student-t copula treatment and Euler allocation for capital carving across major business units.
- ▶ Helped to automate and quicken several manual processes used by the team.
- ▶ Generated ideas for handling then yet uncovered operational risk items including Cyber-risk as well as Climate-risk.
- ▶ Built FFT-based convolution tools using NumPy and SciPy for joint-distribution modelling, and Python distribution-fitting tools for arbitrary empirical distributions.
- ▶ Created COVID-19 R0 estimation models using polynomial regression and automated enterprise risk templates for emerging-risk capture.
- ▶ Studied the impacts of cyber risk, climate risk and legislative risk within operational-risk assessment contexts.

 Python / NumPy / SciPy / Pandas

 Excel / VBA

 C++ Libraries

Head of Market Risk Model Validation Maybank Group

Mar 2014 – Mar 2019

- ▶ Led a regional team of five model validators across Singapore and Malaysia, validating traded and non-traded portfolio valuation and risk models.
- ▶ Validated multi-asset-class instrument and derivative pricing, valuation and risk models across FX, IR, equity, commodity, credit, liquidity and IRRBB domains.
- ▶ Developed independent pricing and validation engines using Python, C++/XLL, VBA, R and SQL-connected data sources.
- ▶ Authored model documentation standards, workflow policies, validation thresholds and governance frameworks, and presented findings to senior management, auditors and banking supervisors.

 C++ / XLL

 Python / SciPy / NumPy / pyxll

 VBA

 SQL

Senior Quantitative Analyst Ambank Group

Apr 2009 – Mar 2014

- ▶ Developed benchmark pricing models, implied and (Dupire-Gatheral) local FX volatility-surface and Swaption volatility-cube builders, correlation tools and derivative analytics for FX, equity and interest-rate products.
- ▶ Implemented practical quant utilities in C++, Python and VBA for valuation review, sensitivities, market-data treatment and model behaviour analysis.
- ▶ Implemented parallel risk models for validation of Murex calibration and pricing engines as well as black-box 3rd party engines like Bloomberg et al - for plain vanilla derivatives to full structured products with complex payout structures.
- ▶ Supported market-risk methodology and quant coaching, translating derivative concepts into maintainable tools for day-to-day risk and validation workflows.

 C++

 Python / early NumPy

 VBA

Risk Specialist Pilot Multimedia

Jun 2007 – Jan 2008

- ▶ Built XML staging tools and VB/VBA plug-ins for a proprietary C# credit-risk application, connecting formula definitions to executable model components.
- ▶ Implemented logistic-regression based probability-of-default utilities and supported translation of credit-risk concepts into software workflows.

 VB6 / VBA

 C#

 XML

Market Risk Analyst Fortis Bank (BNP Paribas Fortis)

Aug 2006 – Jan 2007

- Developed Asian Basket option pricing and Greeks models in C++ and Python, combining analytical approximation, FFT and Monte Carlo methods.
- Implemented and compared PCA-Basket, Deelstra's Basket Bounds, Curran 2M+, Curran, Ju, Gram-Charlier and simulation-based techniques to understand pricing behaviour and sensitivity profiles, involving Eigenvalue decomposition, Monte Carlo, Stochastic Taylor expansion, cylindrical correction, Gaussian Quadrature and other advanced mathematical methods.

</> C++

</> Python / NumPy / FFT

Contracts Analyst Electrabel SA (ENGIE)

Aug 2000 – May 2006

- Developed spark-spread and crack-spread pricers in VBA/C++ for energy derivative workflows, connecting contract economics to valuation and risk treatment.
- Implemented hourly profiling tool using Winter's method in SAS (using macros and IML), GARCH-VaR, FFT-based Carr-Madan styled option pricing methods, Weibull-Dirac Ogive pricing tool for spark spread challenger model
- Implemented a rudimentary markov-based Efficient Frontier approach for discussion purposes
- Assisted the R&D and Econometrics team with Database and SQL management as well as MS Office automation of database-spreadsheet analytic tooling and reporting - forming the basis of the first dashboards used by the Middle Office team
- Supported energy derivatives integration into Murex Front-Office module, with potential flow through to the back-office

</> VBA

</> C++

SAS

 Murex



Education

MBA, Global Management Hochschule Bremen Sep 1999 | Grade: 1.0

BA, Business Administration University of Hertfordshire, UK

Mar 1998 | 2nd Upper Honours



Certifications

- AICB – Risk Management: Principles & Framework
- AICB – Risk Models, Capital & ALM
- Microsoft Visual Basic 6

Hobby Projects

ExcelBridge

Zero-install spreadsheet analytics connected to server-side Python numerical computing.
github.com/geneboo/gene_products/releases/tag/v2.2.8

Fit_it!

Distribution fitting and KDE platform for empirical data analysis.
fit-it-app.run.app

FunkyGraffy

Interactive mathematical visualization and edutainment platform.
funkygraffy.onrender.com

FourierDoodler

Browser-based Fourier epicycle drawing and animation tool.

WebGL Audio Visualizer

Three.js/Web Audio GPU-accelerated reactive visual system.

Large Parquet Viewer

Rust/Polars-style tooling for large parquet datasets with SQL/PRQL queries and local LLM NL-to-query experiments.

SaaS Universe

Mini REST endpoints for derivative pricing, quant analytics, Kalman filters and forecasting tools.

Written Works

Risk-Neutral Pricing via MGFs under Non-Normality

Moment-generating-function drift correction under risk-neutral valuation.

[HTML](#) [PDF](#)

Mind the Gap (Under the Curve)

Numerical integration, quadrature and financial engineering applications.

[HTML](#) [PDF](#)

Understanding Fourier Transform

Fourier foundations, convolution-based option pricing and Python implementation.

[PDF](#)

FFT Cross Correlation vs. Pearson

FFT-based correlation methods compared with Pearson correlation.

[HTML](#) [PDF](#)

Art of Buy-Side Arbitrage

Quantitative perspective on buy-side arbitrage operations.

[HTML](#) [PDF](#)

CUDA / Matrix Multiplication Notes

CUDA tiling and high-performance numerical computing explorations.

Linguistics

English – Native

German – Intermediate

Flemish – Basic +

Malay – Basic

Mandarin – Fluent

French – Basic +

Cantonese – Basic +

Cloud, AI & Creative Computing

Cloud / API Deployment

Google Cloud Run, serverless deployment, REST APIs, Postman, Grafana, Kibana and lightweight analytics services.

AI / LLM Tooling

Applied ChatGPT, Microsoft Copilot, Gemini, Grok, DeepSeek, Kimi, Ollama, Hugging Face models, Cursor and VS Code AI tooling for coding, documentation, debugging and analytics workflow prototyping; integrated local models as NL-to-PRQL/SQL translators for a parquet-viewer concept.

Creative Systems

WebGL, Three.js, SVG, Canvas, audio-reactive visuals, Fourier epicycles, DSP experiments and C++/Rust/JUCE learning projects.